

ARTICLE

Animal Ecology

Habitat associations of six-lined racerunners in longleaf pine managed with a short fire rotation for northern bobwhites

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Abstract

The longleaf pine (*Pinus palustris*) savanna ecosystem is an imperiled, fire-dominated community that supports exceptionally high levels of species richness and endemism. Area of this community has declined by more than 95% due to unsustainable logging, fire suppression, and changes in land-use practices. In recent decades, efforts to restore fire-dominated communities like longleaf pine savanna have gained popularity, especially in light of benefits to charismatic species like the northern bobwhite (*Colinus virginianus*). Although reptiles are important members of this ecological community, far less information exists as to how this group responds to longleaf pine management, especially when game bird conservation is a primary management focus. Although bobwhite management in these systems is mostly synonymous with longleaf pine restoration, additional conservation practices aimed at game birds (promoting fallow fields, supplemental feeding, meso-carnivore control, cross sectional mowing, etc.) might affect the extent to which squamates benefit from habitat management. To better understand how squamate reptiles may benefit from longleaf pine savanna managed for northern bobwhites, we surveyed for six-lined racerunners (*Aspidoscelis sexlineata*) across a large, contiguous tract of longleaf pine with varied land cover characteristics, managed to maximize the conservation of northern bobwhites. Racerunner detection probability on transect surveys was low ($\hat{p} = 0.23$); however, occupancy probability was relatively high ($\hat{\psi} = 0.60$) across the property and driven by percent open ground (positive; 25-m scale), percent grass cover (negative; 25-m scale), and percent wetland (negative; 100-m scale). Our findings support those of past studies about six-lined racerunners in longleaf pine savannas suggesting the species thrives in the context of a short fire rotation (e.g., 2–3 years), even when game bird management is a primary objective of conservation efforts. Racerunners may also specialize on microhabitats (e.g., upland areas with relatively high bare

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ground cover) that occur most frequently in recently burned portions of bobwhite management units.

KEYWORDS

forest management, game birds, prescribed fire, quail, reptiles, squamates

INTRODUCTION

The longleaf pine savanna ecosystem, found throughout parts of the Southeastern United States, is an imperiled community that supports exceptionally high levels of species richness and endemism (Kirkman et al., 2016). This ecosystem, though comprised of many important plant species, is chiefly characterized by longleaf pine (*Pinus palustris*) and understory grasses like wiregrass (*Aristida stricta*; Peet & Allard, 1993). Once, 36 million ha (90 million acres), the area of longleaf pine ecosystem, has been reduced by more than 95% due to unsustainable logging and land conversion (Ware et al., 1993). Efforts to restore longleaf pine systems have begun to gain traction (e.g., Natural Resources Conservation Service [NRCS] Longleaf Pine Initiative), but individual longleaf pine trees grow slowly and are highly dependent upon “frequent fire,” which means that this community can only persist when supported by deliberate management (i.e., prescribed burning; Glitzenstein et al., 1995; Palmquist et al., 2015). In fact, the presence of frequent fire is also essential for the conservation of many associated plant species (e.g., Venus flytrap [*Dionaea muscipula*], yellow pitcher plant [*Sarracenia flava*]) and animal species (southern fox squirrel [*Sciurus niger niger*], red-cockaded woodpecker [*Leuconotopicus borealis*]; Brewer, 2006; Means, 2006; Outcalt, 2000).

Longleaf pine savannas with frequent fire (e.g., 2–3 year frequency) host a high species diversity, with at least 46 species of reptiles found in this ecosystem (Means, 2006). For example, the gopher tortoise [*Gopherus polyphemus*] is a large, burrowing turtle that creates essential microhabitat for a wide array of pine-land species and, as such, is often regarded as a key-stone species (Catano & Jack Stout, 2015). In fact, gopher tortoise burrows not only provide microhabitats for many species, but they also provide critical refugia for many taxa to use during the frequent fire events that characterize this ecosystem. Longleaf pine savannas host high reptile species diversity with some of those species being habitat specialists (eastern indigo snake [*Drymarchon couperi*], pinesnake [*Pituophis melanoleucus*], etc.; Guyer & Bailey, 1993; Steen et al., 2013; Van Lear et al., 2005). With that in mind, one of the most promising opportunities to conserve native reptiles is through management efforts

targeted at other conservation-reliant species. For example, management efforts aimed at species like the red-cockaded woodpecker and northern bobwhite (*Colinus virginianus*) use fire to stifle woody understory growth within longleaf pine systems and might benefit associated reptile communities (Van Lear et al., 2005; Wahlenberg, 1946). Management for northern bobwhites, specifically, also often includes secondary practices like fallow field establishment/promotion (Richardson et al., 2020), supplemental feeding (Henry et al., 2021), cross sectional mowing (Malone et al., 2022; Sinnott et al., 2023), and meso-carnivore control (Palmer et al., 2019). While these secondary conservation practices are demonstrated to benefit game birds, they are known to alter snake behavior and/or abundance as well as the ecology of exotic species like red imported fire ants (*Solenopsis invicta*; Palmer et al., 2019). Therefore, while it is possible that the conservation of “high profile” vertebrates (like bobwhites) in longleaf pine savannas may also provide quality habitat for native reptiles, few studies have occurred in this context (Howze & Smith, 2022; Steen, 2011; Steen et al., 2013).

To better understand how squamate reptiles may benefit from longleaf pine savanna managed using a short fire rotation (2 year) for a high-profile game bird, we surveyed for six-lined racerunners (*Aspidoscelis sexlineata*) across a large tract of longleaf pine managed to maximize northern bobwhite abundance. Specifically, we (1) estimated six-lined racerunner detection probability and occupancy rates in longleaf pine-dominated habitats managed with fire for northern bobwhites and (2) assessed factors driving variation in detection probability and occupancy rates. We discuss our results in the context of prescribed fire as a tool to manage longleaf pine and the intersection of game bird/native reptile conservation.

METHODS

Study area and sampling locations

We studied six-lined racerunner ecology in Brunswick County, North Carolina. The study area falls into the southeastern Coastal Plain, an area characterized by a

subtropical climate, and nutrient-poor, well-drained soils (Peet & Allard, 1993). Within Brunswick County, we surveyed a 1850-ha contiguous stand of longleaf pine savanna managed for hunted game species, especially northern bobwhites. Although the dominant land cover was mature longleaf pine savanna, there were other cover types naturally interspersed: wetlands, early successional longleaf pine, dirt roads/firebreaks, and open water (ponds, bays, etc.). There were also fallow fields throughout the property (usually 1–2 ha in size) that were sown with partridge-pea (*Chamaecrista fasciculata*) and/or ragweed (*Ambrosia artemisiifolia*) to support northern bobwhite brood-rearing. In addition to fallow field cultivation, the property implemented other conservation practices to support quail: supplemental feeding (0.26 cubic meters per hectare per year [3 bushels per acre per year] of wheat and sorghum), year-round meso-carnivore trapping, and prescribed fire with a rotation cycle of every 2 years (half the property burned each year).

Six-lined racerunner surveys

To establish sampling locations, we gridded the study area into 100 cells with a sampling point plotted in the center of each. This approach generated sampling locations that were each ~400 m from its nearest neighbor. We used this systematic sampling approach to maximize coverage of the study area while maintaining the spatial independence of survey locations. Each of these 100 sampling locations became the centroid for a visual transect for six-lined racerunners. We conducted repeat surveys along 50-m transects, oriented south to north, centered over the survey points, from 3 July through 22 August, 2023, between 0630 and 2300. We visited each of the 100 points three times in random order (100 sites $\times$ 3 visits = 300 total surveys). We did not restrict surveys to particular weather conditions within this window because we were interested in understanding the factors that might explain variation in detection probability. Prior to each survey, we recorded survey covariates: cloud cover (estimated to the nearest 25%), precipitation (none, mist, light rain, heavy rain), temperature, Beaufort Wind Index, ordinal date, and time of day. We walked each transect for two minutes (i.e., 25 m per minute) and recorded whether six-lined racerunners were detected during each survey (1/0). Because we visited each site three times, we generated a detection history of 1 and 0s that served as our response data for occupancy models (MacKenzie et al., 2002), described below.

Vegetation surveys

We sampled local vegetation (within 25 m) of each transect (hereafter, “microhabitat”) once between 30 July and 12 August. We estimated percent cover of vegetation strata and vegetation within 25 m of each site using the same transect as surveyed for racerunners, sampled with an ocular tube (James & Shugart, 1970) and a Nudds board (Nudds, 1977), respectively. We estimated percent cover using an ocular tube at 10 “stops” per site, one at each of the following distances along the transect: 5, 10, 15, 20, and 25 m, for north and south. At each “stop,” we recorded bare ground, leaf litter, coarse woody debris, moss, grass, forb, shrub, fern, sapling, and canopy. Bare ground was defined as un-vegetated soil. Leaf litter was defined as any organic debris smaller than 10 cm. Coarse woody debris was any organic debris larger than 10 cm in diameter. We considered moss to be any live, nonvascular plant. Grasses were any graminoid. Forbs were any broad-leaved dicotyledon. We defined a shrub as a woody plant with stems branching above ground. Ferns were seedless vascular plants. Saplings were any tree less than 10 cm in dbh; trees larger than 10 cm in dbh were considered canopy trees (McNeil et al., 2018). Finally, we assessed vegetation density via a Nudds Board (2 m tall with 20 squares, each 20 $\times$ 20 cm in size), placed at point center. We read the Nudds board from 10 m north and south of the board from a height of 1 m. Each reading consisted of a visual assessment of the number of squares (out of 20 possible) obscured at least halfway by vegetation. We averaged the two Nudds board readings at each site and the number of “hits” for each vegetation stratum (/10 stops) for percent cover of each microhabitat stratum.

Remote-sensed data

To assess habitat at a broader spatial extent, we extracted metrics from an unpublished land cover map developed in QGIS (QGIS Development Team, 2022). Relevant land cover values included (1) mature pine, (2) open grassland (including grassy areas with short, sapling pines), (3) wetland, (4) brood field, and (5) open water. We generated rasters that depicted “distance to nearest...” for each of these covariates with the raster package in R (Hijmans, 2018). We also generated distance to nearest road. In addition to these “distance to nearest...” variables, we calculated the “percent cover within 100 m” for each variable (except “road” because it was a line vector file). Finally, we extracted the normalized difference vegetation index (NDVI) at each point using a raster developed for the property in 2019, obtained from the property manager.

Analyses

We created single-season occupancy models in the R package unmarked (Fiske & Chandler, 2011; MacKenzie et al., 2002; R Core Team, 2023) to assess factors influencing six-lined racerunner occupancy. We only specified univariate models in this study (no additive covariates) to avoid over-parameterizing our models. We began by fitting a “detection” model set where we held occupancy probability constant and specified models with each of the following survey covariates: temperature at the time of survey, ordinal date, Beaufort Wind Index, minutes since sunrise, cloud cover, and two habitat covariates that might explain observers’ ability to see racerunners: NDVI and vegetation density. We also included quadratic terms for temperature and minutes since sunrise because these variables might exhibit a nonlinear trend. We also included a null (intercept only) detection model for comparison: $p(\cdot), \psi(\cdot)$ in all model sets. We ranked all models in descending order of Akaike information criterion adjusted for small sample size (AIC_c ; Akaike, 1998; Burnham & Anderson, 2002). We also assessed the biological importance of covariates by examining whether β 95% CIs overlapped zero, interpreting those that did so as having weak effects on detection/occupancy (Arnold, 2010). After identifying a top detection model, we incorporated its detection terms into all of our following occupancy models. This second and final model set included univariate models for each of the microhabitat variables and each of the landscape variables. Finally, we also assessed model goodness-of-fit by calculating $\hat{c}$ via the MacKenzie and Bailey goodness-of-fit test (Kéry & Royle, 2020; MacKenzie & Bailey, 2004). If our top model was overdispersed ($\hat{c} > 1.0$), we adjusted our model ranking using quasi- AIC_c ($QAIC_c$).

RESULTS

We conducted 300 surveys at 100 locations (3 visits each) and, during these, made 66 six-lined racerunner observations across 33 of the locations (naïve occupancy probability = 0.33). Structural characteristics measured in the field varied among sites. More specifically, sites usually included a small amount of bare ground (median: 10%), abundant leaf litter (median: 70%), irregular patches of grass and forbs (median: 30% and 40% respectively), occasional woody cover (median shrub cover: 10%) and a relatively open canopy, typical of longleaf pine savannas (median canopy cover: 10%). At a broader scale, sites were mostly within mature upland pine (median: 54% within 100 m), had a small amount of wetland/drain within 100 m (median: 9%), and were a

median distance of 18 m from a dirt road. Our null detection model ($p(\cdot), \psi(\cdot)$) indicated that detection probability was, on average, $\hat{p} = 0.23$ (95% CI: 0.13–0.38). This detection estimate would equate to a cumulative detection probability of 0.54 across three visits. After accounting for imperfect detection, mean racerunner occupancy probability was estimated at $\hat{\psi} = 0.60$.

Our top occupancy model was slightly overdispersed ($\hat{c} = 1.38$), so we ranked all models using $QAIC_c$. In the detection-only model set, there were five competing models (Table 1). Because one of these models was the intercept-only null model, we fit our occupancy models with no detection covariates. In our occupancy covariate model set, there were four competing models: (1) $\psi(\text{NDVI})$, (2) $\psi(\% \text{ wetland within 100 m})$, (3) $\psi(\% \text{ grass within 25 m})$, and (4) $\psi(\% \text{ bare ground within 25 m})$; Table 2). The β 95% CIs on NDVI overlapped zero, so we interpreted its biological effect as weak. Percent wetland within 100 m had a negative relationship with racerunner occupancy and sites with >22% wetland had below-average occupancy probability (Figure 1A). Percent bare ground cover within 25 m had a positive relationship with racerunner occupancy and sites with <13% bare ground cover had below-average occupancy probability (Figure 1B). Finally, percent grass cover within 25 m had a negative relationship with racerunner occupancy and sites with >47% grass cover had below-average occupancy probability (Figure 1C). Collectively, these results suggest that longleaf pine sites most likely to support racerunners have little grass, patches of bare ground, and very little wetland cover (Figure 1D,E).

TABLE 1 Single-season occupancy models where occupancy probability (ψ) was held constant ($\cdot$) and detection probability (p) varied as a function of survey covariates.

Model name	k	$\Delta QAIC_c$	w	Quasi-LL
$p(\text{temperature}), \psi(\cdot)$	4	0.00	0.25	−84.98
$p(\text{ordinal}), \psi(\cdot)$	4	0.16	0.23	−85.06
$p(\text{wind}), \psi(\cdot)$	4	1.45	0.12	−85.70
$p(\cdot), \psi(\cdot)$	3	1.67	0.11	−86.90
$p(\text{vegetation density}), \psi(\cdot)$	4	1.85	0.10	−85.90
$p(\text{time of day}), \psi(\cdot)$	3	1.96	0.10	−85.96
$p(\text{cloud cover}), \psi(\cdot)$	4	3.76	0.04	−86.86
$p(\text{NDVI}), \psi(\cdot)$	4	3.79	0.04	−86.87
$p(\text{temperature}^2), \psi(\cdot)$	5	8.20	0.00	−87.97
$p(\text{time of day}^2), \psi(\cdot)$	5	19.85	0.00	−93.79

Note: Shown for each model is the model name, number of model parameters (k), Δ quasi Akaike information criterion adjusted for small sample size ($QAIC_c$), model weight (w), and the quasi-log likelihood (Quasi-LL). Model components with “ $\cdot$ ” in the terms indicates no covariates used (i.e., intercept only).

TABLE 2 Single-season occupancy models where occupancy probability (ψ) varied as a function of survey covariates.

Model name	k	ΔQAIC_c	w	Quasi-LL	β coefficients, β_1
$p(\cdot), \psi(\text{NDVI})$	4	0.00	0.23	−83.42	−19.70
$p(\cdot), \psi(\% \text{ wetland } 100 \text{ m})$	4	0.50	0.18	−83.67	−4.10
$p(\cdot), \psi(\% \text{ grass } 25 \text{ m})$	4	1.61	0.10	−84.22	−1.41
$p(\cdot), \psi(\% \text{ bare ground } 25 \text{ m})$	4	1.66	0.10	−84.25	0.08
$p(\cdot), \psi(\% \text{ water } 25 \text{ m})$	4	2.07	0.08	−84.45	...
$p(\cdot), \psi(\% \text{ canopy } 25 \text{ m})$	4	2.35	0.07	−84.59	...
$p(\cdot), \psi(\text{distance to road})$	4	3.05	0.05	−84.94	...
$p(\cdot), \psi(\text{cover type})$	5	3.50	0.04	−84.06	...
$p(\cdot), \psi(\% \text{ mature pine } 100 \text{ m})$	4	4.42	0.02	−85.63	...
$p(\cdot), \psi(\cdot)$	3	4.78	0.02	−86.90	...

Note: Shown for each model is the model name, number of model parameters (k), Δ quasi Akaike information criterion adjusted for small sample size (ΔQAIC_c), model weight (w), and the quasi-log likelihood (Quasi-LL). For competing models ($\Delta\text{QAIC}_c < 2.0$), we also show the β coefficient on model covariates and indicate whether their 95% CIs overlapped zero (italics) or did not overlap zero (boldface). Variables on ψ included normalized difference vegetation index (NDVI), cover type, and percent cover of habitat conditions measured at the 100-m radius (remote sensed) or 25-m radius (field-measured). Model components with “.” in the terms indicates no covariates used (i.e., intercept only).

DISCUSSION

Here, we present the first study to focus on patterns of occupancy for six-lined racerunners in a longleaf pine savannah managed specifically for northern bobwhites. Increasingly, efforts like the USDA-NRCS “Working Lands For Wildlife” program (Litvaitis et al., 2021) and “Bobwhites, Grasslands, and Savannas” program have gained momentum as conservationists increasingly aim to manage habitat for northern bobwhites (WLFW, 2018). Although bobwhites are commonly recognized as an umbrella species for many taxa (Carroll et al., 2017; Kamps et al., 2017), few have assessed whether reptiles are “covered” by the “bobwhite umbrella” (Guyer & Bailey, 1993; Ruthven et al., 2008; Steen et al., 2013). Our study demonstrates that such sites managed for northern bobwhites can also provide quality habitat for reptiles like the six-lined racerunner. This appears to be because, while northern bobwhites ultimately require a “mosaic” of characteristics (to support nesting, brood-rearing, nonbreeding cover, etc.; Palmer & Sisson, 2017), some of these characteristics appear to overlap with the needs of six-lined racerunners (Figure 1). Should racerunner occupancy be a primary focus of a particular management effort, conservationists could likely increase occupancy even beyond what we observed here by avoiding wetlands and ensuring low grass cover with abundant open patches (Figure 1, Table 2). Still, prescribed fire, when used to conserve bobwhites, may benefit squamate reptiles, even if racerunners’ needs are not explicitly considered (Ashton & Knipps, 2011; Duarte et al., 2017; Mushinsky, 1985). This is an important advancement in our understanding of reptile conservation because

secondary conservation practices (e.g., meso-carnivore control) could affect racerunner ecology in potentially unpredictable ways by increasing snake and fire ant predation pressure (Palmer et al., 2019). Beyond racerunners, we anecdotally observed other regionally rare species during surveys including northern pine snakes, chicken turtles (*Deirochelys reticularia*), and oak toads (*Anaxyrus quercicus*), many of which benefit (or are expected to benefit) from fire-focused longleaf pine management efforts (Guyer & Bailey, 1993).

One challenging aspect of our study was that racerunner detection probability was low on visual transects ($\hat{p} = 0.23$). In fact, even over three visits, cumulative detection probability was only 0.54, indicating that the species was commonly undetected. Further, to achieve a cumulative detection probability of 0.95 (McNeil et al., 2014), 12 visits would be required ($1 - (1 - 0.23)^{12} = 0.96$). This highlights the need to account for low detection probability, as we have here, by implementing both “design” based accounts (e.g., surveying in warm, dry conditions during the summer) and “model” based accounts (e.g., analytical approaches that model detection failure; MacKenzie et al., 2002). Alternative analytical approaches that might be useful for this application would be distance models (Buckland et al., 2015) or N-mixture models (Kéry & Royle, 2020). From a study design perspective, there may be additional ways to bolster detection probability; for instance, only sampling in the hottest part of the day (Winne & Keck, 2004). Although our “null” detection model was competing (Table 1), a model with temperature at the time of survey suggested a positive effect with β 95% CIs that did not overlap zero. Similar findings have

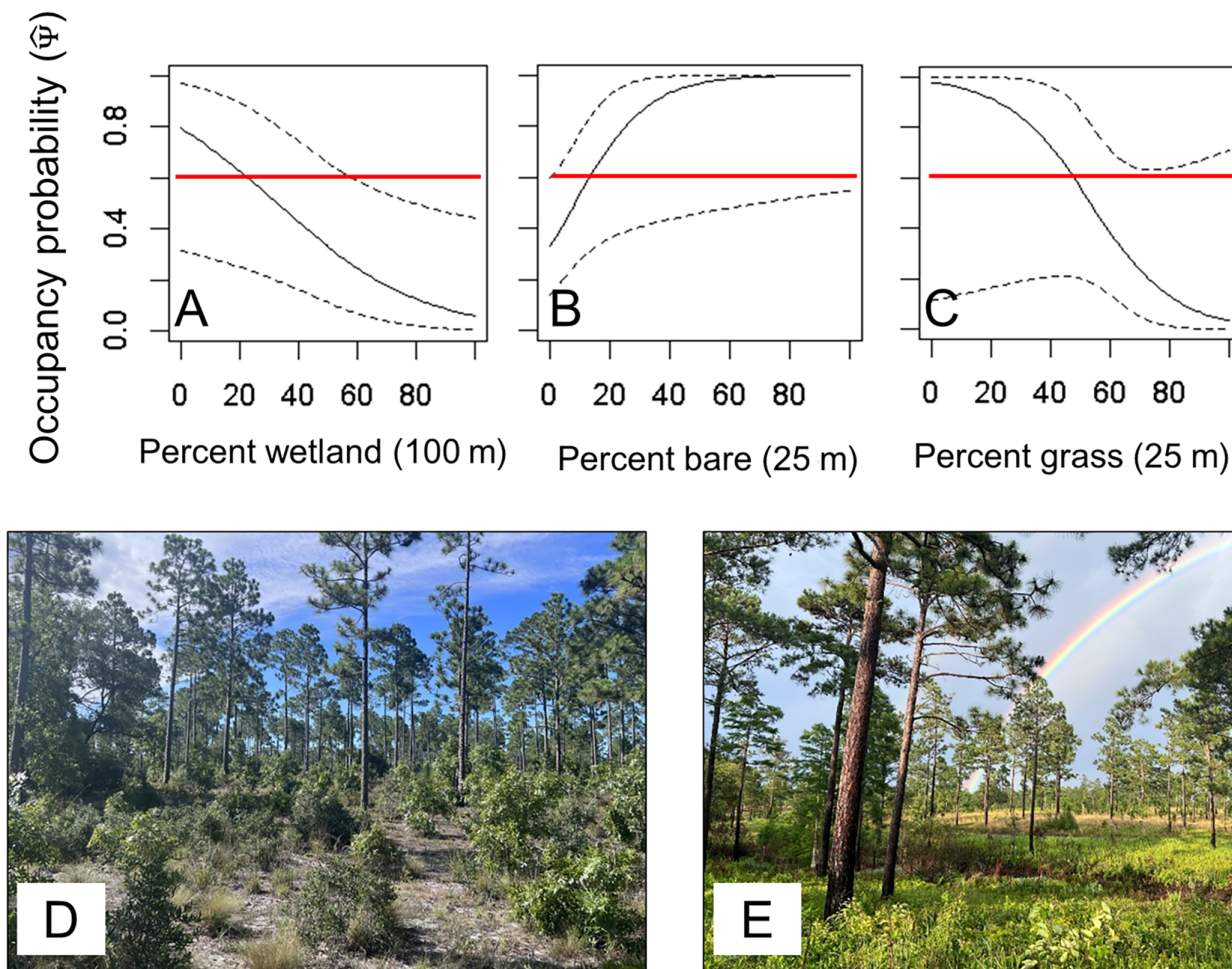


FIGURE 1 Our top occupancy models (Δ quasi Akaike information criterion adjusted for small sample size [Δ QAIC_c] <2.0) indicated that six-lined racerunners (*Aspidoscelis sexlineata*) were most likely to occupy sites that had (A) very little wetland cover within 100 m, (B) high bare ground cover within 25 m, and (C) little grass cover within 25 m. Solid black lines represent estimated occupancy probability ($\hat{\Psi}$) and dashed lines represent 95% CIs. Red lines represent mean $\hat{\Psi}$ (0.60). Collectively, our results suggest that longleaf pine sites that would support racerunners would have little grass, patches of bare ground, and very little wetland cover, resembling the site in panel (D) and *not* resembling panel (E) (photo credit: Darin McNeil).

been reported by other research that six-lined racerunners are most active when temperatures exceed 38°C (Paulissen, 1988a; Winne & Keck, 2004); however, anecdotal findings contrary to this have been found as well (Bellis, 1964). We specifically sampled in a variety of conditions to potentially capture predictors of detection probability; however, our analyses revealed none. With this finding in mind, racerunner detection probability could be improved by using other survey techniques like drift fences or mark-recapture ($\hat{p}$ = 0.80–0.95; Baumgardt et al., 2021), but such approaches would likely sacrifice the number of sites surveyed due to greatly increased sampling effort required for such methods (MacKenzie et al., 2002).

In contrast to detection probability, occupancy probability was surprisingly high ($\hat{\Psi}$ = 0.60; Figure 1) and nonrandom with respect to habitat covariates (Table 2); more specifically, we found wetlands and local grass

cover to have a negative relationship with occupancy while bare ground to exhibit the opposite pattern. Six-lined racerunners are not known to have an affinity for wetlands; in fact, studies have shown that the species is drawn to very dry sites with well-drained soils (Baxley & Qualls, 2009; Burt, 1931; Fitch, 1958; Howze & Smith, 2022; Paulissen, 1988b), supporting our findings. Likewise, bare ground cover likely affords opportunities for racerunners to thermoregulate (Paulissen, 1988a), potentially while simultaneously searching for prey. Six-lined racerunners are active predators that rely heavily upon the capacity to detect prey via scent and sight, which may also explain this pattern (Paulissen, 1986; Pehl, 2021). Finally, our observation that racerunners prefer microhabitat conditions with limited grass cover is also consistent with the species' primary hunting tactic: the species' captures prey primarily through active pursuit (Fitch, 1958; Paulissen, 1986). It seems likely that dense grass and other

vegetation might inhibit racerunners' capacity to pursue prey and escape predation.

Given widespread declines in the longleaf pine savanna ecosystem over the last century, conservation of species like the northern bobwhite is widely viewed as a conservation "win" (Dailey & Crosby, 2015; Weber et al., 2022). Our results provide the first clear evidence that conservation of northern bobwhites and longleaf pine savanna can support to the persistence of disturbance-dependent reptile species like the six-lined racerunner. We also provide some potential strategies that could enhance this system for racerunners—particularly incorporating fire that promotes open ground and low grass cover, especially away from wetlands. Most historic longleaf pine forests have been lost due to land conversion (Brockway, 2005; Outcalt, 1996; Ware et al., 1993), and it appears that the northern bobwhite can act as an umbrella species for, at least, this reptile. Previous studies have examined six-lined racerunner use of longleaf pine savanna and found, as ours did, that the species responds positively to the microhabitat conditions created by frequent fire (open canopies and bare ground) and can occur at relatively high densities (Dodd et al., 2007; Howze & Smith, 2022; Steen et al., 2013). Studies have also examined the effects of fire on the six-lined racerunner outside of longleaf pine savanna and have found increased abundance in response to burning (Ashton & Knipps, 2011; Lindsay et al., 2023). However, one drawback of our study was that we used occupancy as a primary response; studies that examine demographic rates across a suite of habitat characteristics would likely be more informative (Van Horne, 1983). Moreover, a before-after-control-impact (or similar) study design aimed at assessing the true response of racerunners to bobwhite management which would be valuable for understanding how un-managed sites are improved by this conservation focus. Regardless, we believe that this study provides a useful first assessment of the factors driving reptile use patterns within longleaf pine savanna managed with frequent fire for game birds.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

DATA AVAILABILITY STATEMENT

Data (Samek et al., 2024) are available from Dryad: <https://doi.org/10.5061/dryad.cc2fqz6gk>.

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